Arslan Kenbayev

+82 10 7921 7883 | kenbaevarslan@gmail.com | linkedin.com/in/arslan-kenbayev | github.com/arskenchik | arskenchik.github.io

Education

KAIST — Aug. 2022 – Feb. 2027
Bachelor of Science in Electrical Engineering — *Daejeon, South Korea*

National School of Physics and Math — Sep. 2018 – May 2022
High School, Math and Physics, GPA: 3.95/4.00 — *Almaty, Kazakhstan*

Experience

Research Assistant — Aug. 2025 – Present
QAIST — *Almaty, Kazakhstan*
- Advancing wildfire detection by developing sub-mW neuromorphic IoT sensor nodes for field-deployed wireless mesh networks.
- Reduced integration risk by writing system-level electrical interface specs and coordinating hardware/software requirements with KAIST researchers and industry partners.

Embedded Systems Engineer Intern — May 2025 – Mar. 2026
AXInvent — *Daejeon, South Korea*
- Accelerated smart agriculture prototyping by selecting components, designing I2C/SPI/UART sensor schematics and power rails.
- Designed, integrated and validated analog/digital embedded hardware components on prototyped PCBs with power regulation and sensor signal conditioning.
- Validated field IoT sensor data integrity through oscilloscope/DMM bring-up, sensor debugging, and edge-to-cloud integration across Arduino Nano, Nvidia Jetson, Raspberry Pi 4, and server via gRPC/HTTP/2.

Semiconductor Engineer Intern — May 2025 – Aug. 2025
KAIST IOEL Lab — *Daejeon, South Korea*
- Achieved electroluminescence in organic OLEDs by fabricating CO2-derived/organosilicon thin-film stacks and characterizing performance through forward-bias I-V sweeps.
- Improved OLED optimization workflow by tuning layer composition from I-V data toward lower turn-on voltage and higher luminance for healthcare organic electronics.

Automation Engineer Intern — July 2024 – Sep. 2024
AAEngineering — *Almaty, Kazakhstan*
- Improved gold ore segregation readiness by validating sensor integration in SolidWorks, analyzing multi-node RF/IoT topology, and prototyping automated control logic.

Software Engineer Intern — Feb. 2024 – June 2024
KAIST CIS Lab — *Daejeon, South Korea*
- Extended KakaoMap route planning by implementing the CCH algorithm in C++ to improve dynamic road-network query performance under Prof. Heejin Ahn.

Projects

Pipelined RISC-V CPU | *System Verilog, Verilog, ModelSim* | [github] — Sep. – Nov. 2025
- Designed and verified 100% testbench pass rate for a synthesizable 5-stage RISC-V pipeline CPU by implementing hazard detection/forwarding logic in SystemVerilog and debugging ModelSim waveform timing faults across IF/ID/EX/MEM/WB stages.

Ultra-Low-Power Smartwatch Firmware | *C, Arduino Nano 33 BLE, nRF5 SDK*| [github] — Mar. – June 2025
- Reduced smartwatch average current draw to 85 μA (DMM-verified) by implementing IMU duty cycling, OLED auto-sleep, and coordinated I2C sensor polling across LSM6DS step counter, APDS-9960 gesture sensor, and SSD1306 display, with BLE GATT server firmware on nRF52840 for client synchronization.

Digital Clock | *Verilog, EDA Playground* — Feb. – June 2024
- Designed and verified synchronous digital logic modules including FSMs, flip-flops, counters, sequential logic and MUXes in System Verilog/Verilog using simulation waveform analysis.

Capstone: Industrial Anomaly Detection | *Python, DCASE 2020, Git, Linux* — Mar. – June 2025
- Enabled mechanical failure detection under real-world noise by training an unsupervised autoencoder on ToyADMOS/MIMII acoustic datasets and applying reconstruction error thresholding.

Technical Skills

Hardware & Embedded: PCB Design, Soldering, Firmware Dev (nRF5 SDK, Segger Studio), Sensor Integration, I2C/SPI/UART/BLE, I-V Characterization, Oscilloscope/DMM
Languages: SystemVerilog, Verilog, C, C++, Python, MATLAB, R, Java
Platforms: Arduino Nano 33 BLE, Nvidia Jetson, Raspberry Pi, RISC-V
EDA / CAD: Fusion 360, SolidWorks, ModelSim, EDA Playground
Developer Tools: Git, VS Code, Linux